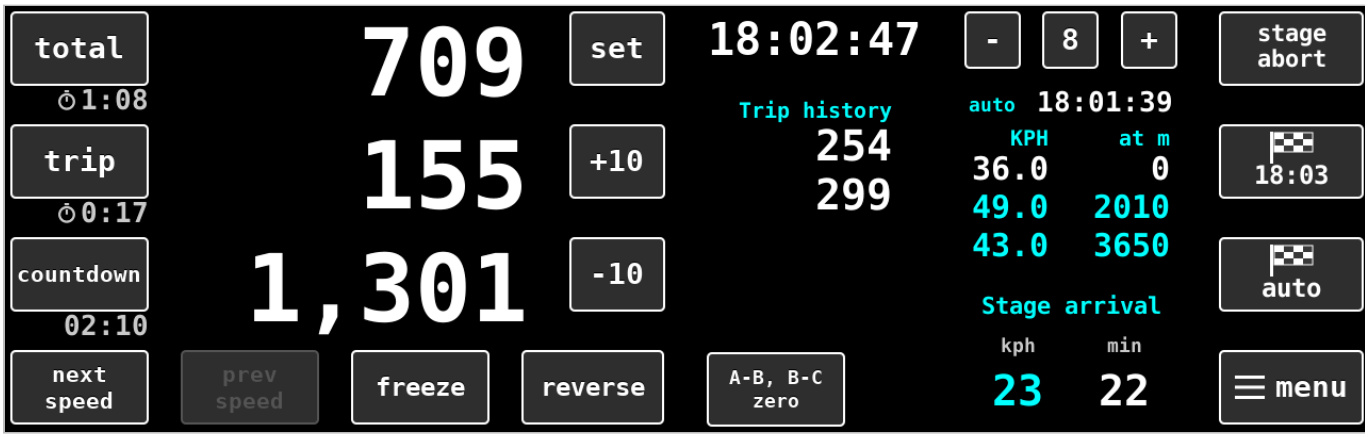
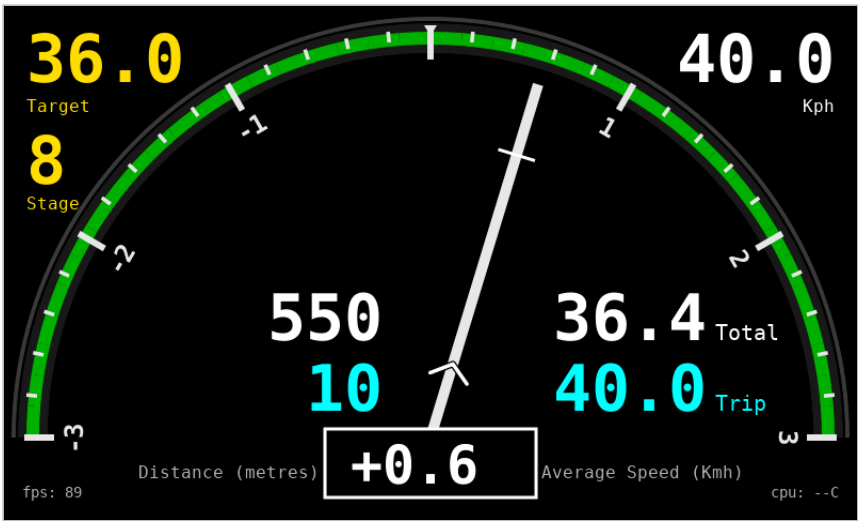
Rallybox User Manual

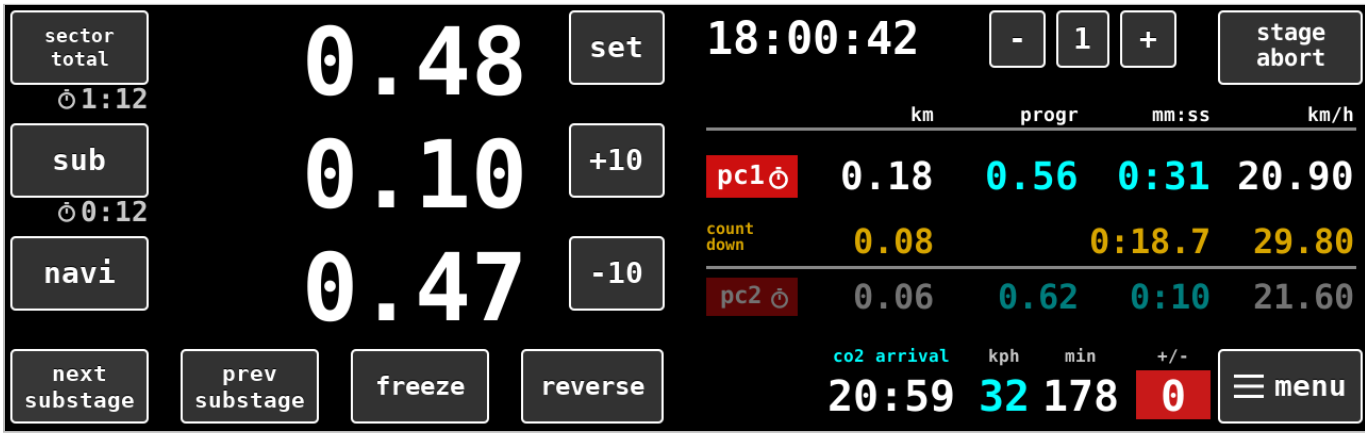
Software Nero2.1 · manual updated 30 October 2026



Co-pilot display, navigation page (greek rally type, stage running).



Driver display.



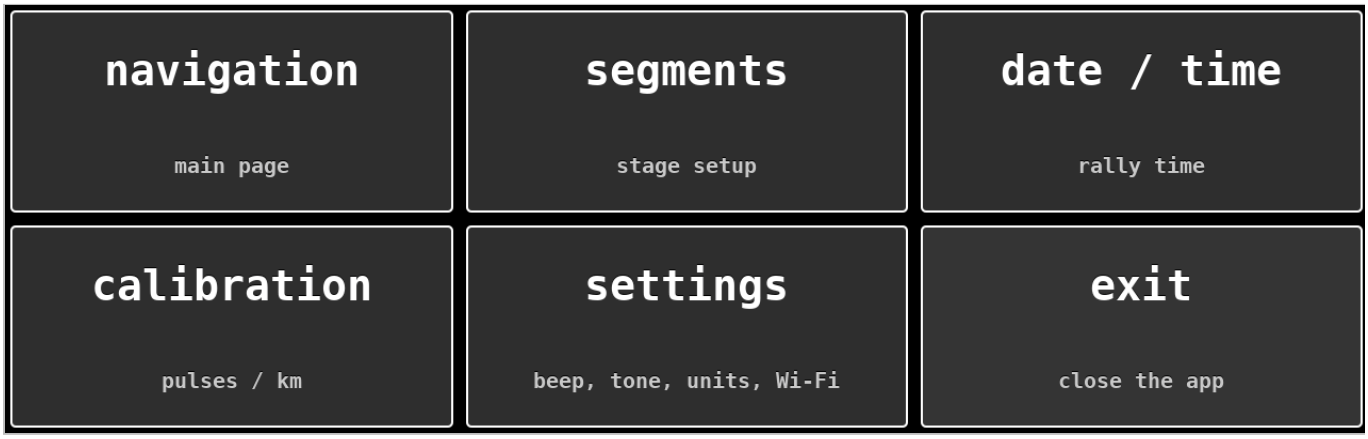
Co-pilot display, navigation page (italian rally type, inside a timed trial).

RALLY TYPES

The box works in one of three **rally types**, chosen under `settings` › **Rally type**. The choice changes the navigation and segments pages.

Rally type	Use it for	In this manual
greek	Stages start at roadbook boxes with zero distance at a set minute; speeds change at roadbook boxes.	Programming, Starting, Navigating
italian	A stage is a **sector** from one time control (CO) to the next, made of untimed driving, timed trials (pc), average trials (pm), stamp checks (ct) and the closing time control (co).	Programming, Starting, Navigating
surrey	Stages start at landmarks (with unspecified distances) within roadbook boxes displaying cumulative distance; scored on time taken.	Programming, Starting, Navigating

GETTING AROUND



The menu page.

Every page has a `menu` key at the bottom right. Press it to reach the main menu.

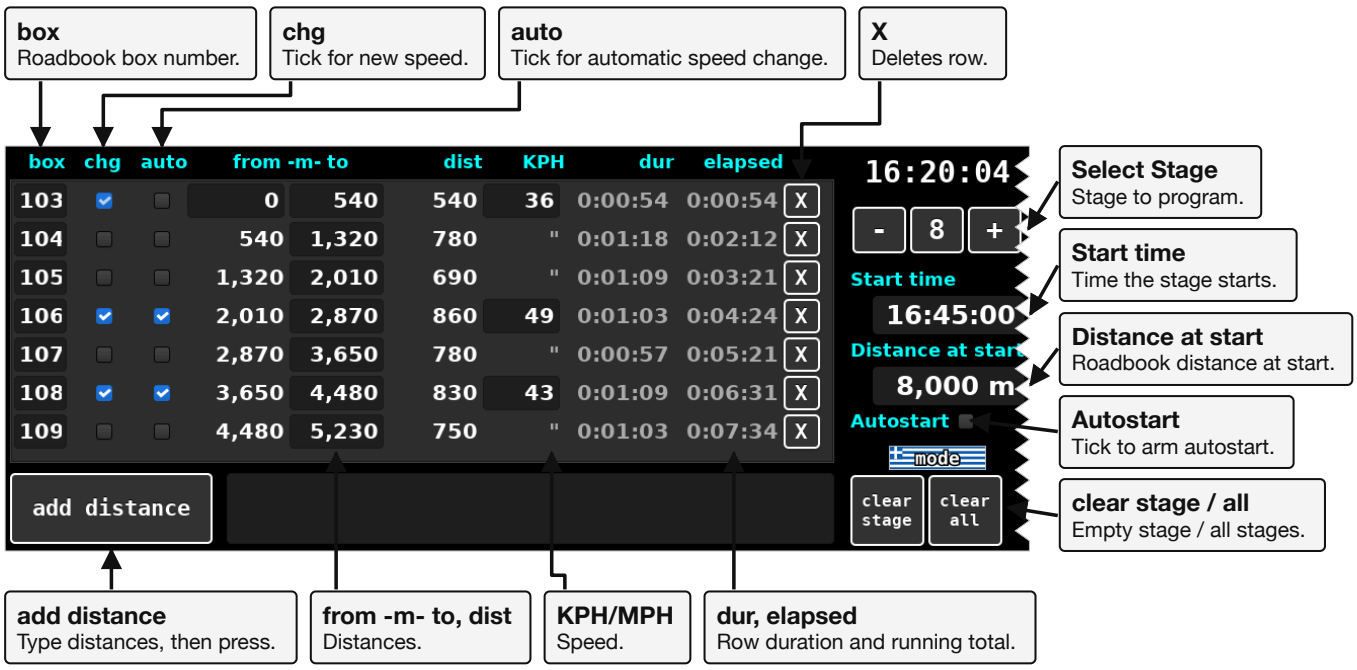
- **navigation** – used while navigating and on stages.
- **segments** – stage setup.
- **date / time** – set rally time.
- **calibration** – set pulses per kilometre.
- **settings** – beep, tone, rally type, units and logging; the spanner key opens the **Setup** sub menu (displays, Bluetooth speaker, Wi-Fi and the phone link, software updates).
- **exit** – saves the settings and closes the program. It does not switch the Raspberry Pi off.

Numbers are typed on the on-screen keypad: C clears a selected entry and the arrow key deletes one character. There is no Enter key: an entry is saved when you tap another box or leave it alone for a second.

GREEK RALLY TYPE

Programming Greek Stages

Select **greek** rally from the settings page and then select `segments` from the main menu. The segments page will display a Greek flag. The system defaults to metres (rather than km) but this can be overridden in settings.



Stage 8: one row per roadbook box, speed changes at boxes 103, 106 and 108.

Entering Greek stage data

Enter every roadbook box in the stage, or only the rows with speed changes. With every box entered, Beep Assist navigation (when on) beeps ahead of each box.

1. Select the stage with **– n +**, or tap the number (1–99).
2. Tap the box beside `add distance` and type the cumulative distance at which each row ends, separated by semicolons: `540;1320;2010` (or `0.54;1.32;2.01` in km). Later entries are appended or inserted as needed.
3. Press `add distance`.
4. For rows with speed changes tick **chg** and enter the speed.
5. Tick **auto** for the box to change speed automatically at the specified distance. If unticked, the navigator must press `next speed` on the navigation page.
6. **Start time** – when the stage starts (hh:mm:ss).
7. **Distance at start** – the roadbook distance at the start line; used for Stage arrival.
8. **Autostart** – tick to arm an autostart at today's start time (a past time is refused). It can also be switched on/off on the navigation page.

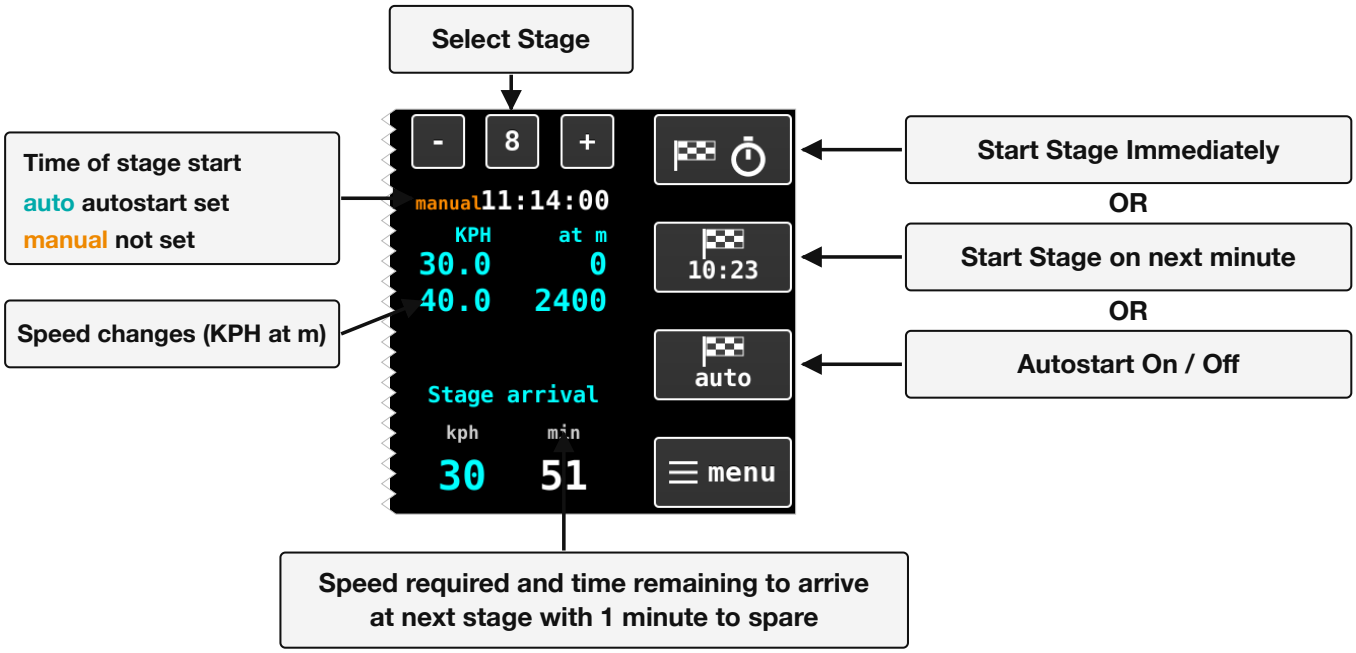
Entries save automatically under the stage number shown. A running stage cannot be edited.

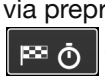
`upload stage` – currently only available on phone app.

Clearing stages

`clear stage` empties the stage shown; `clear all` empties every stage. Both ask first.

Starting a Greek Stage



A stage can be started via preprogrammed autostart (using the time programmed in the Segments section), or by pressing the Start Now  or Start On Next Minute  button.

Autostart: When autostart is armed the button  will appear as black on white, and the word **auto** (cyan) appears next to the start time. Autostart can be quickly toggled on/off using the Autostart button on the navigation display. When Autostart is not armed the word **manual** (orange) appears next to the start time. A start time that has passed, with nothing armed or running, is crossed out in red.

What happens with autostart:

- With less than 5 minutes to the start the driver's display will show a countdown clock.
- The Total and Trip distances will zero at the start time.
- The timer resets at the start time.

Start on Next Minute: The Start on Next Minute button constantly updates to show the next whole minute. Pressing it sets a type of autostart at that minute.

What happens with Start on Next Minute:

- Pressing the button automatically overrides any programmed autostart.
- The Total and Trip distances zero immediately (not at the departure minute). Any distance driven between pressing and the start minute will be included in the stage distance.
- The timer resets at the start minute.

Start Now: Pressing it starts a stage immediately.

What happens with Start Now:

- Pressing the button overrides any programmed autostart.
- The Total and Trip distances zero immediately.
- The timer resets immediately.

Starting scenarios

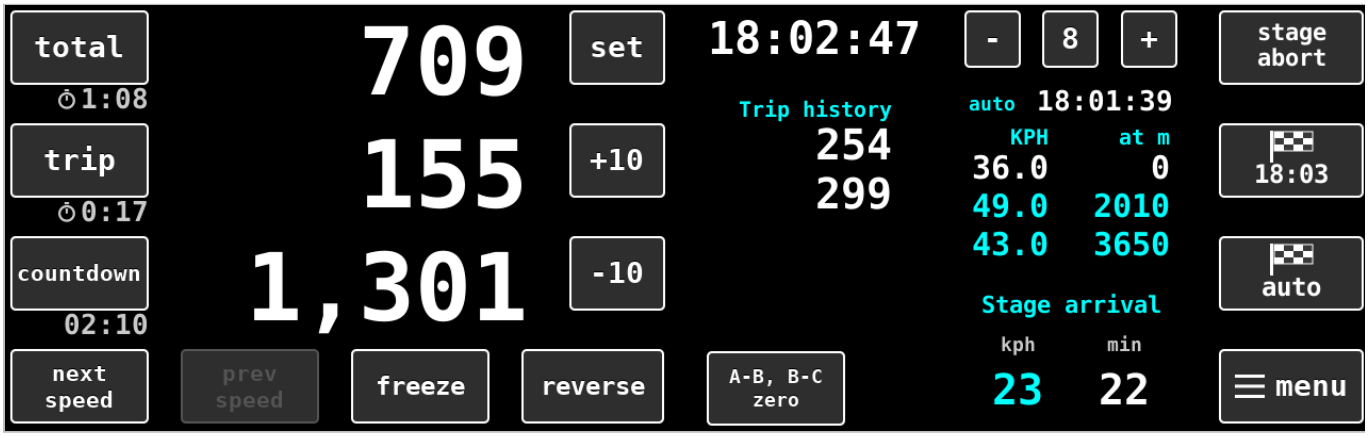
Regular on-schedule start – in most instances this will be handled correctly by a pre-programmed autostart which resets the distances and timer at the allotted start time.

Late arrival at start – there are 2 ways to handle this scenario.

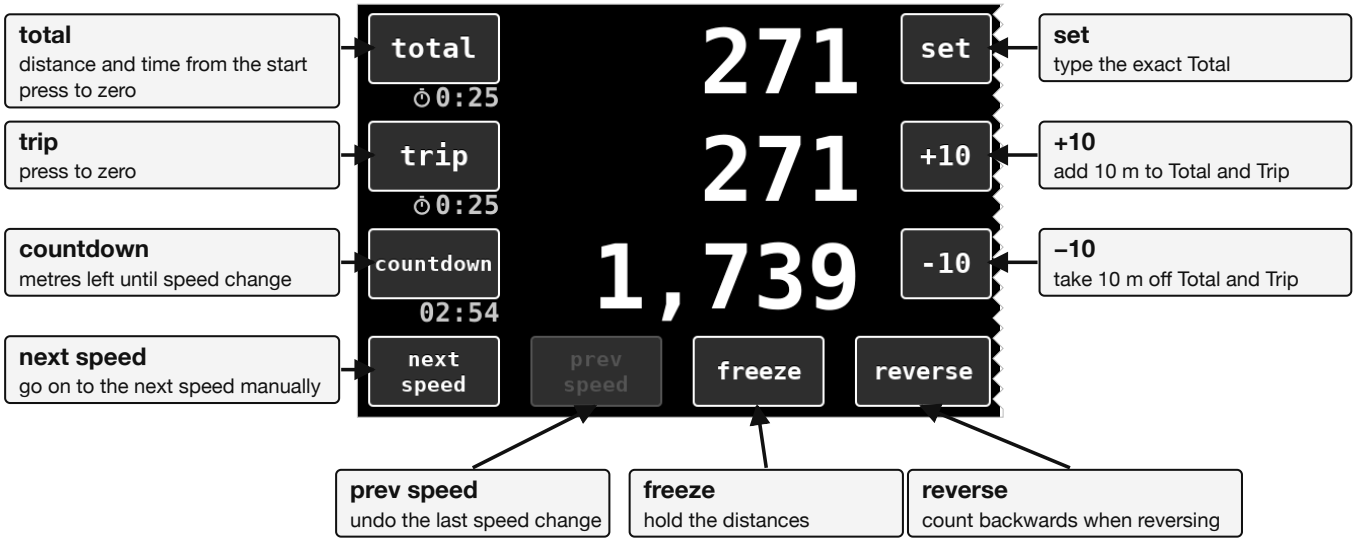
- Have autostart start the stage on time (before you have arrived) and press Total to zero the distances at the start line and attempt to catch up during the stage. This is the preferred method.
- Press Start Now – if a stage is already running the box will ask you to confirm but will measure as of the original button press. Because the timer will have started late the driver will need to catch up and target a reading where the seconds ahead is equal to the delay in starting.

Obstructed start – if traffic or the car may make you late to the first timing point, press the next-minute start key (flag with a time) on the start line instead of using autostart, and move off shortly before departure. Total and Trip reset when pressed; timing starts at the minute.

Navigating a Greek Stage



Stage 8 running, 709 m in: 1,301 m and 2 min 10 s to the speed change at 2010 m. Trip was zeroed twice (Trip history 299, 254); Stage arrival needs 23 kph to reach the next stage with 22 min to go.



total

Distance and elapsed time since the start. Press to zero the distance.

- Outside a stage: zeroes the distance and timer, not Trip.
- Waiting for a timed autostart: zeroes Total, Trip and the timer without asking; the autostart zeroes them again at its start time. (An autostart cannot be armed during a stage.)
- If Start on Next Minute has been selected: zeroes Total and Trip without asking; the timer starts at the minute.
- During a stage (a missed start): asks first, then zeroes Total and Trip from the press; the stage clock keeps running.

trip

Shows intermediate distance and time. Press to zero it; the old figure appears in the **Trip history** column. Trip also restarts at every speed change.

countdown

Distance and time to the next speed change. If that change is not automatic, press `next speed` there.

next speed and prev speed

`next speed` is used to move to the next speed manually. `prev speed` undoes the last speed change. The buttons can also be used to override automated speed changes.

+10, set and –10

`+10` / `–10` move Total and Trip 10 m per press; `set` allows a precise distance to be entered for Total (Trip unchanged). During a stage, corrections apply across average speeds and timing errors.

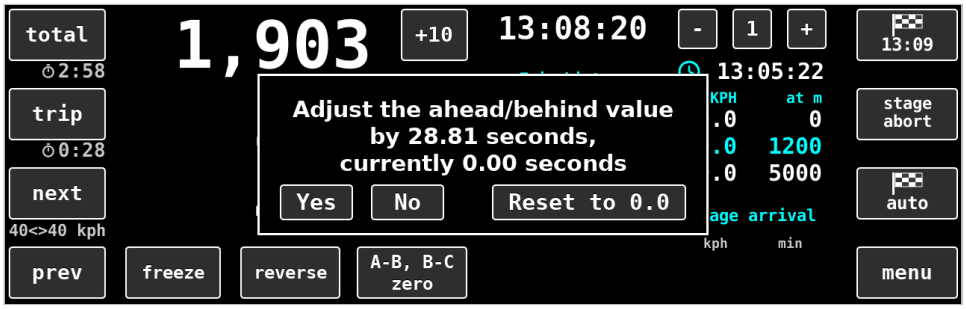
freeze and reverse

`freeze` stops the distances from counting while lit; clocks and timers continue – use it after a navigation error. On release, counting resumes from the frozen figures. `reverse` makes Total and Trip (as well as countdown) count backwards. It is typically used when reversing after overshooting a junction, as the box only counts pulses and has no concept of direction. Freeze overrules Reverse if both are pressed. The buttons display as black on white when active and a whooping sound is emitted.

A-B, B-C zero

Where a rally is measured on an AB, BC basis and this option is selected in settings, the `A–B, B–C zero` button appears under the Trip history. Pressing this during a stage when a timing mark or timing marshal is seen sets the timing error in the driver's display to zero, so that the average speed calculation is made independently between each timing mark.

The popup shows the error to remove and the adjustment already applied.



- `Yes` zeroes the needle at this instant.
- `Reset to 0.0` removes the adjustment so the true error shows.
- `No` changes nothing.

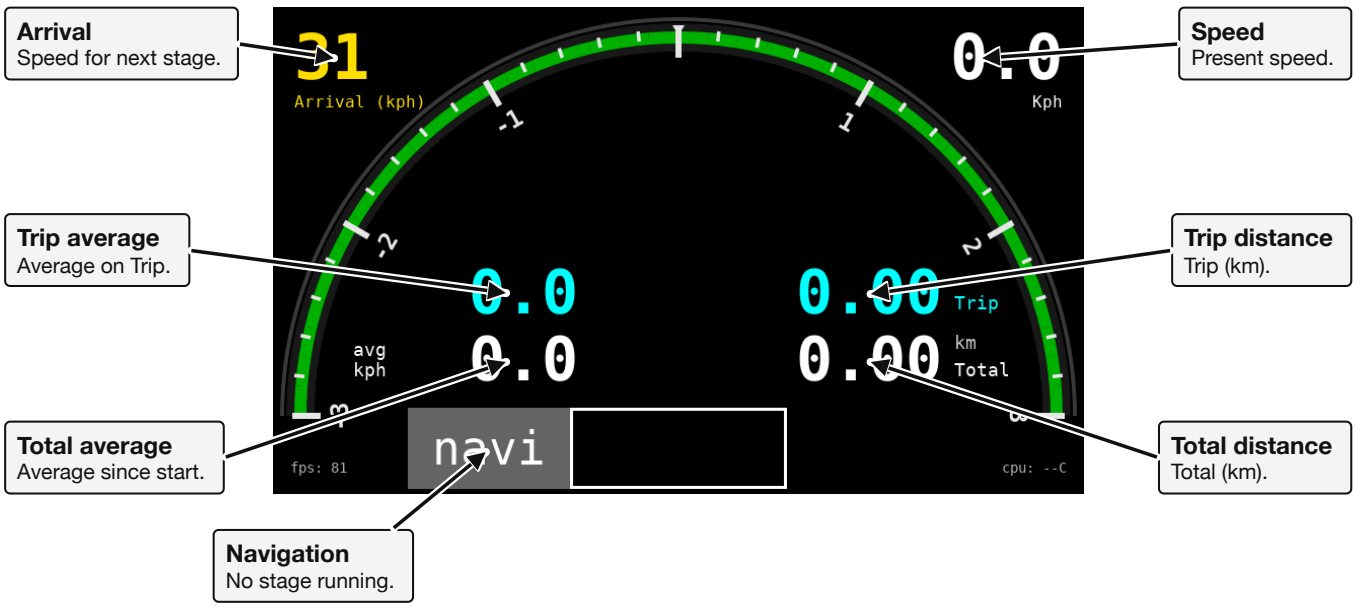
The stage panel display

The column shows the selected stage, the start time of the stage and autostart status, the distances at which speed changes commence and the speed needed (and allotted time) to reach the next stage with 1 minute to spare.

The Driver Display in Greek mode

The driver display has two pages in Greek mode: one between stages (navigation) and one in a stage.

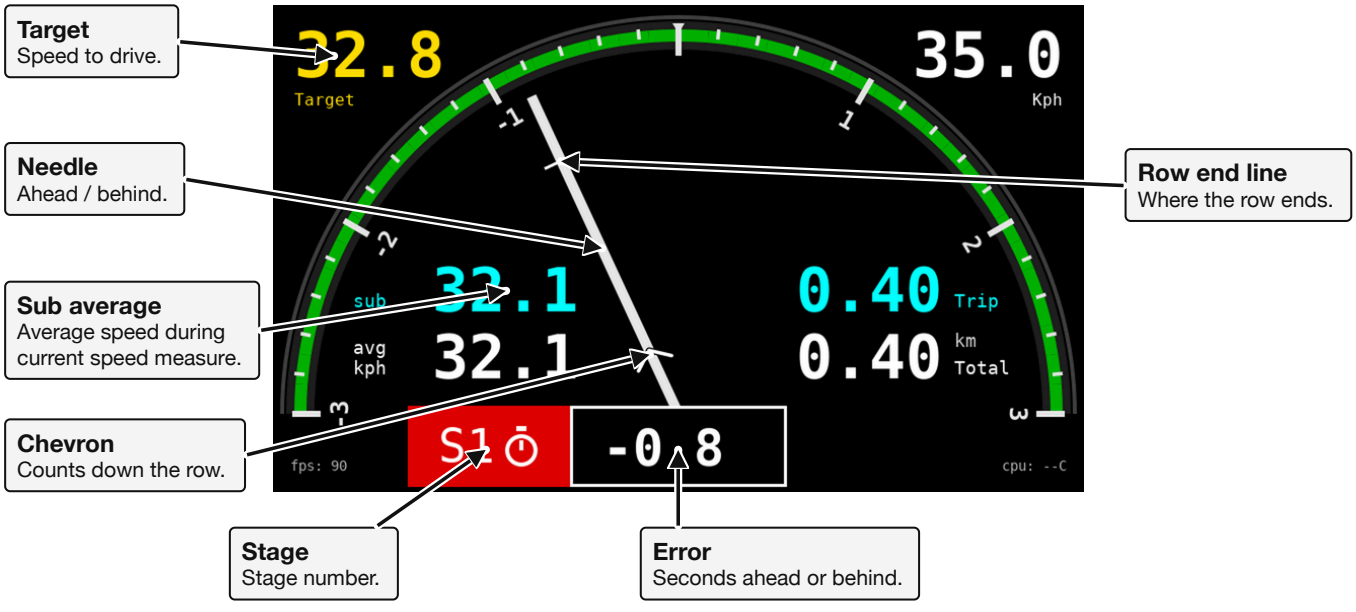
Navigation (between stages)



With no stage running there is no needle. The top left shows the **arrival** speed: the speed needed to reach the next stage with a minute to spare, the same figure as the co-pilot's Stage arrival. The distances and averages are Trip and Total.

In a stage

The stage display differs where marked.



During a timed stage the needle shows how far ahead/behind the driver is in seconds. If the error exceeds the current gauge scale (changeable from 1–3 seconds in settings) the needle will stay pinned far left (or far right if ahead) and the scale will expand. The red logo shows the stage number (S1, S2 …). The cyan **sub** average is your average speed since the current speed section began, that is, the distance driven divided by the time elapsed since the last speed change in the stage data (see Entering Greek stage data). It restarts at each speed change, so it goes with the time error; the white average is the average since the start. The chevron slides up the needle and reaches the **row end line**, the short cross-bar on the needle, as the box ends.

The chevron

One chevron rides the needle. At the start of each row it sits 20% of the way up the needle; as the row is driven it slides up toward the **row end line**, arriving as the row ends, then drops back to the start for the next row. It only counts the row down: it does not show how far ahead or behind you are (the needle does that) and there is only ever one. Between stages there is no needle and no chevron.

While an autostart is pending, a T− `hh:mm:ss` countdown shows mid-display with **Early Departure ENABLED** or **DISABLED** beneath, depending on whether a regular autostart or a start on next minute is triggered.

ITALIAN RALLY TYPE

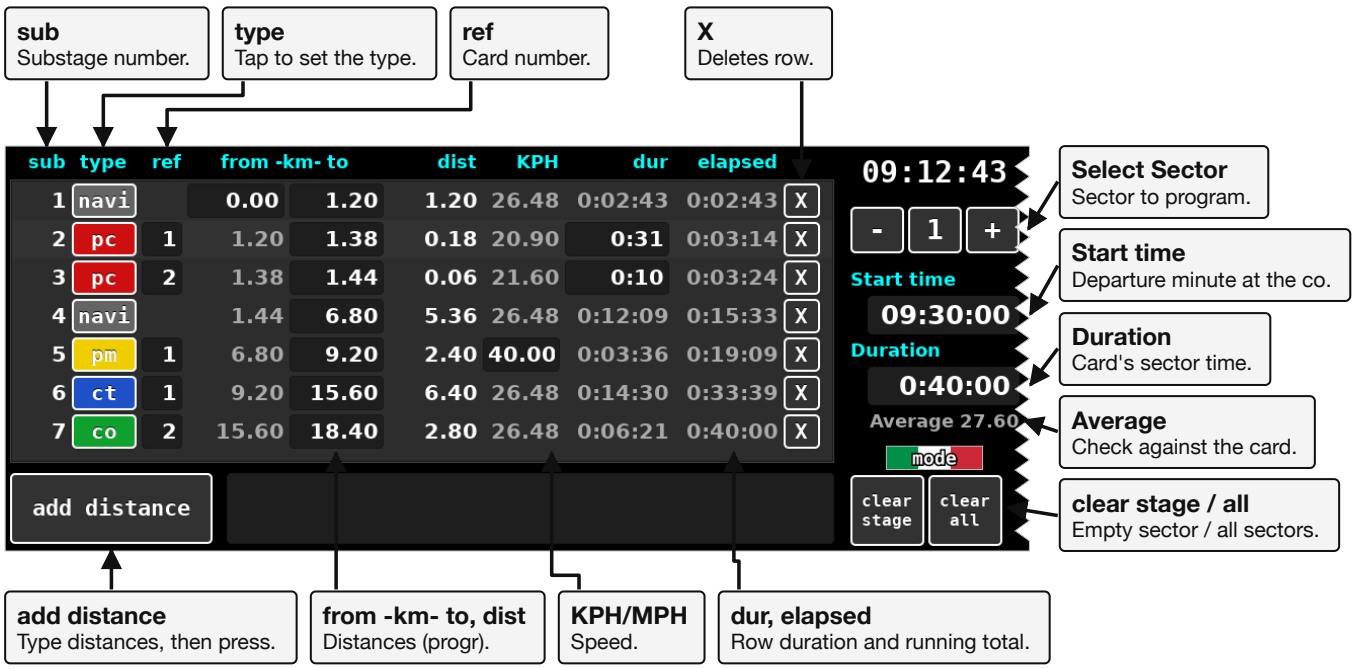
Italian rallies have many short back-to-back timed trials, so each time-control section is programmed as linked substages.

The substage types:

Type	What it is	How the box treats it
navi (grey)	Untimed driving between controls.	A required speed is worked out from the sector's total duration so the car arrives mid-way through its arrival minute.
pc (red, stopwatch)	Timed trial between two pressure tubes. Between linked pcs a single tube ends one pc and starts the next.	Speed is worked out from the subsector's duration.
pm (yellow)	Average speed check with unknown timing points.	Duration is worked out from the given speed.
ct (blue)	Stamp check: no time.	Untimed, like navi.
co (green)	The time control that ends the sector; arrive in your minute.	The last row; the co arrival time is worked out from the card.

Programming Italian Sectors

Select **italian** rally from the settings page and then select `segments` from the main menu. The segments page will display an Italian flag. The system defaults to km (two decimals) but this can be overridden in settings.



An example first sector, co 1 → co 2, with one of each substage type: navi, two linked pcs, navi, pm, ct and the closing co.

Entering Italian sector data

Enter every substage on the organiser's Times and Distance Table (TDT) starting from the row after the opening time control, up to and including the closing one (here co 2). The opening co line (0.00 km) is not entered but its time goes in **Start time**.

1. Select the sector with **– n +**, or tap the number (1–99): the number of its starting time control.
2. Tap the box beside `add distance` and type the cumulative distance (**progr**) at which each substage ends, separated by semicolons: `1.20;1.38;1.44;6.80;9.20;15.60;18.40` (or `1200;1380;1440;6800;9200;15600;18400` in metres). Later entries are appended or inserted as needed; take care inserting a distance inside an existing pc or pm which will get recalculated.
3. Press `add distance`.
4. Tap the **type** tag to cycle navi → pc → ct → pm → co. **ref** is the card's number (a new row gets the last of its type plus one but can be overridden).
5. For pc rows enter the card's time (m:ss) in **dur**; the speed is worked out.
6. For pm rows enter the speed in the KPH/MPH column; the duration is worked out.
7. **Start time** – enter the departure minute at the opening co (hh:mm:ss).
8. **Duration** – enter the card's sector duration (h:mm:ss). The **Average** displayed faintly below should correspond to that given on the TDT sheet.

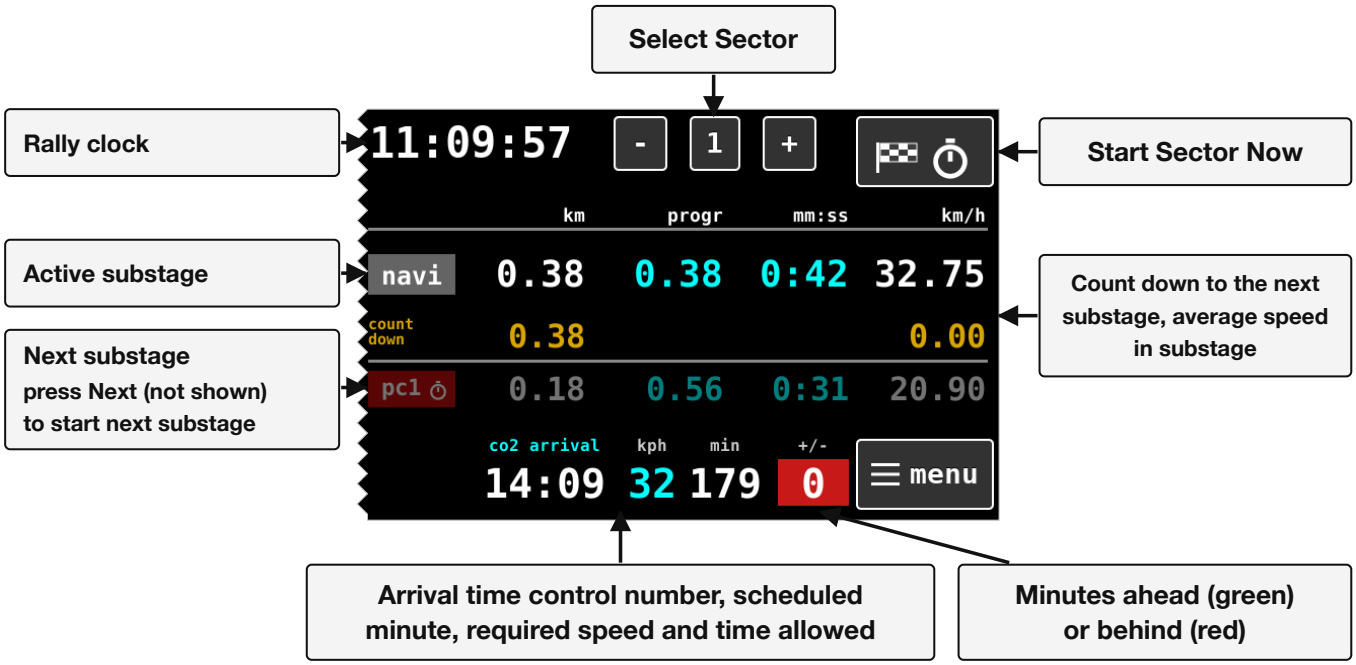
Typed figures are white on dark boxes; worked-out figures grey. Entries are saved automatically. A running sector cannot be edited.

`upload stage` – currently only available on phone app.

Clearing sectors

`clear stage` empties the sector shown; `clear all` empties every sector. Both ask first.

Starting an Italian Sector

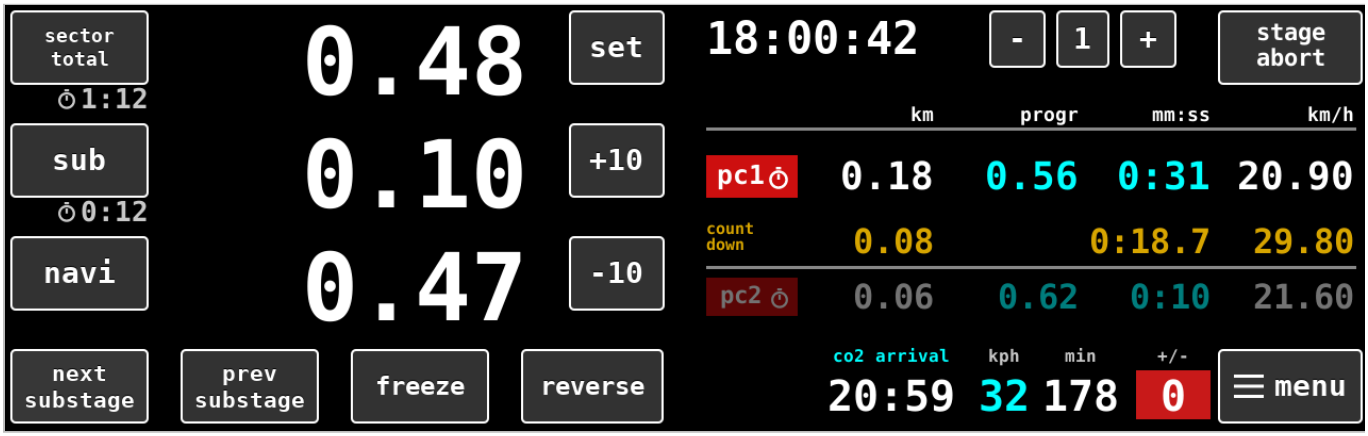


A sector is started at the time control by pressing the start now button  during the **departure minute**. The sector counter counts from that whole minute (11:30:40 counts as 11:30:00) and sets an elapsed time at 30s into the minute, targeting arrival in the middle of the arrival minute. Sector total and sub distances zero; navi distance does not zero at the same time. The `stage now` key reads `stage abort` when a sector is in progress.

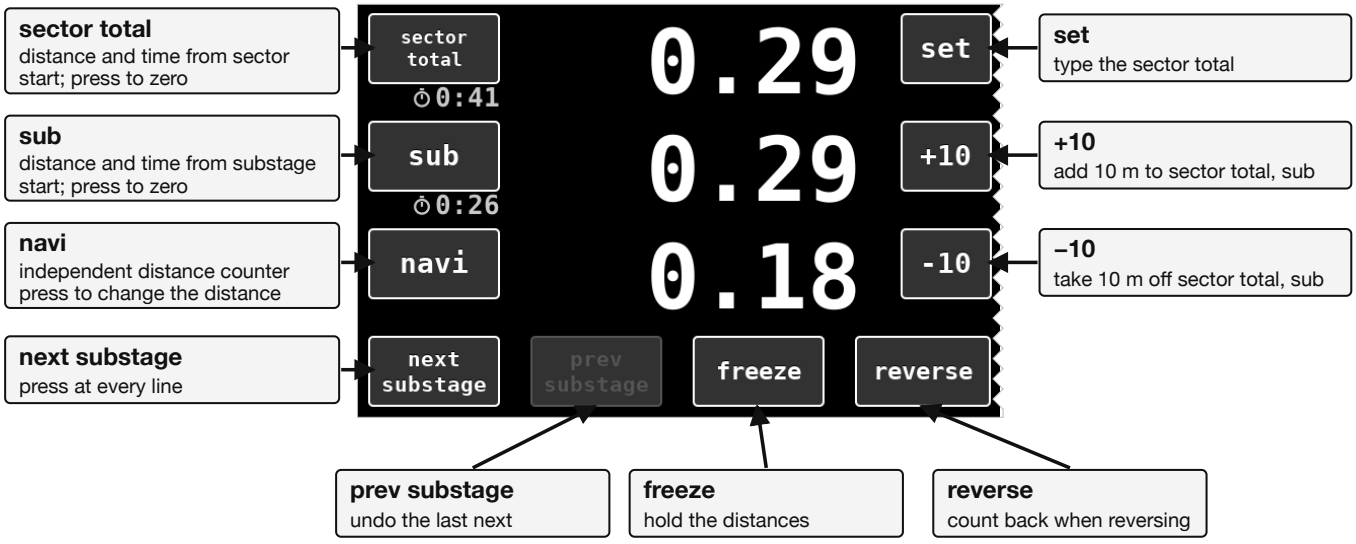
Starting a new substage

- Press `next substage` to enter the next substage (pc, pm etc). NB that this sets the sector total distance counter to the total distance on the card's **progr** entry. The navi distance increments independently – any variation will give a guide to calibration errors as long as Navi is not reset.
- If `next` is pressed early or by mistake press `prev substage` to undo it.

Navigating an Italian Sector



Inside pc 1, 100 m in: 80 m and 18.7 s to its end tube; average so far 29.80.



sector total

Distance and elapsed time in the sector, matching the card's **progr** value. Press to zero.

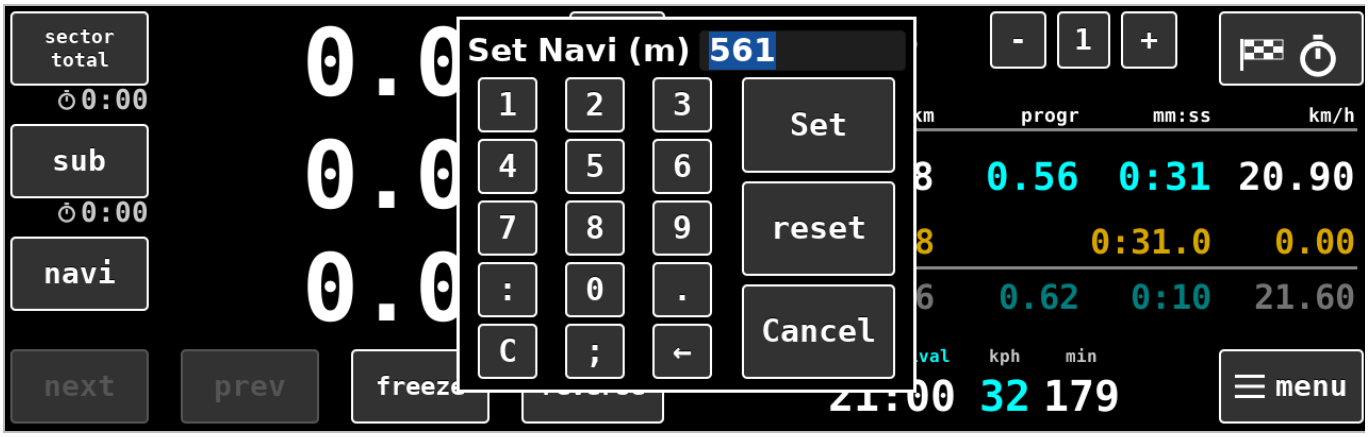
- Outside a sector, it zeroes the distance and the timer but not sub.
- During a sector, it asks first, then zeroes sector total and sub from the press while the sector clock keeps running.

sub

Distance in the current substage; zeroes at every new substage when `next substage` is pressed. Press to zero.

navi

An independent distance counter. Press to change the distance.



Tapping navi: Set Navi.

next substage and prev substage

`next substage` is pressed to move to the next substage. `prev substage` undoes the change.

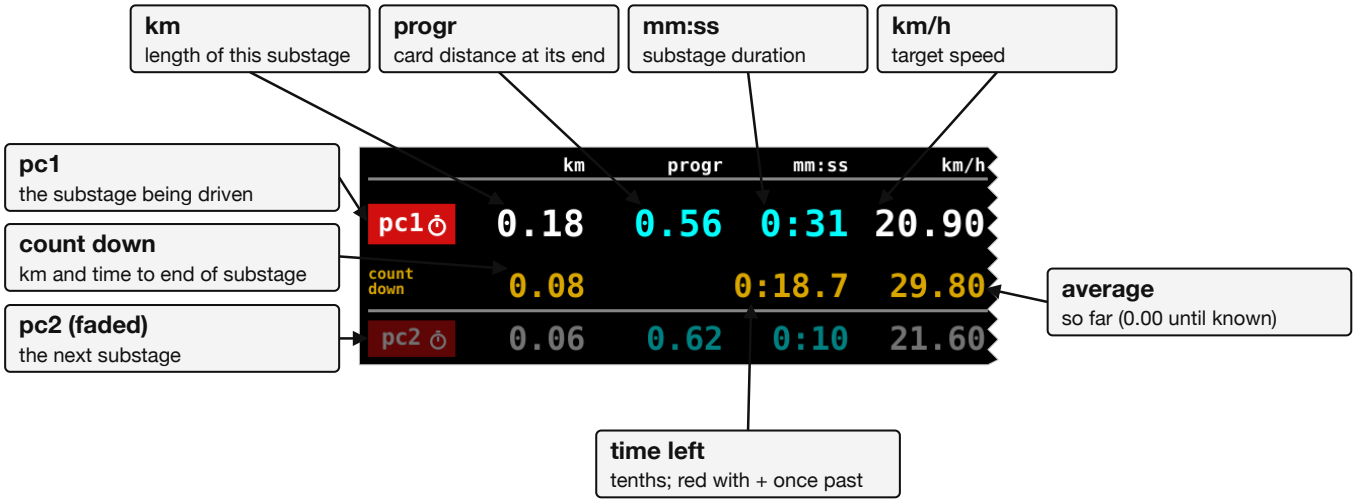
+10, set and –10

`+10` / `–10` move sector total and sub 10 m per press (navi unchanged); `set` allows a precise distance to be entered for sector total.

freeze and reverse

`freeze` stops the distances from counting while lit; clocks and timers continue – use it after a navigation error. On release, counting resumes from the frozen figures. `reverse` makes the distances count backwards. It is typically used when reversing after overshooting a junction, as the box only counts pulses and has no concept of direction. Freeze overrules Reverse if both are pressed. Operates on sector total, sub and navi distances. The buttons display as black on white when active and a whooping sound is emitted.

The substage display

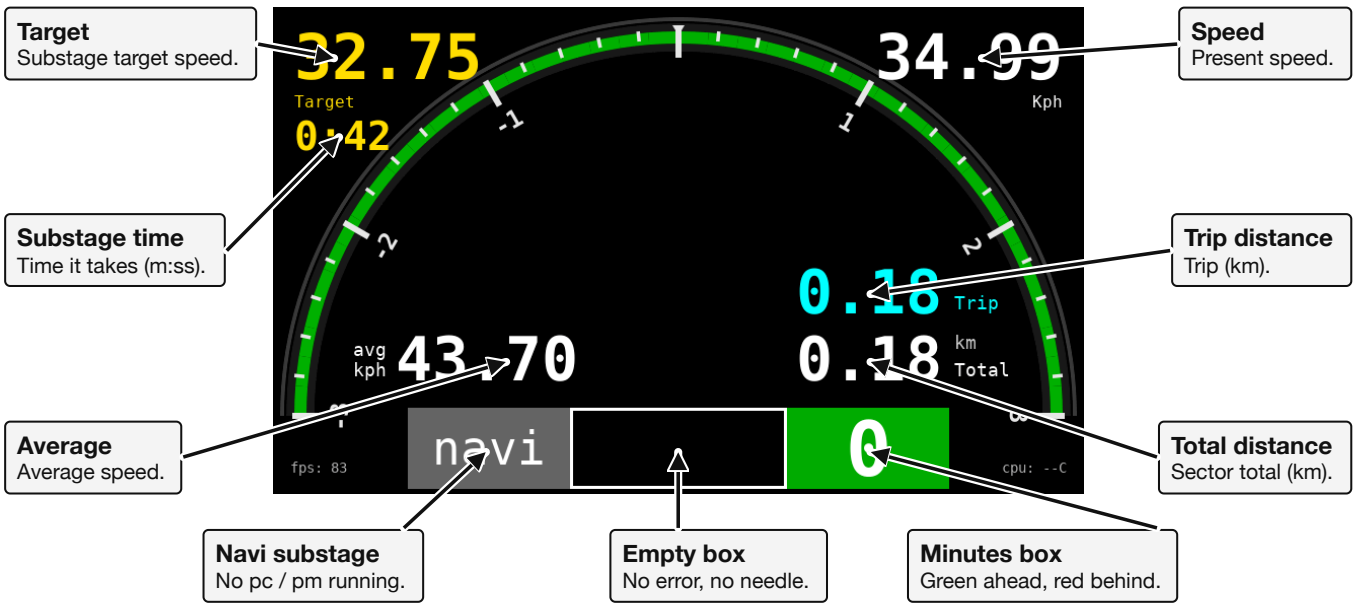


The substage display, inside pc 1.

The Driver Display in Italian mode

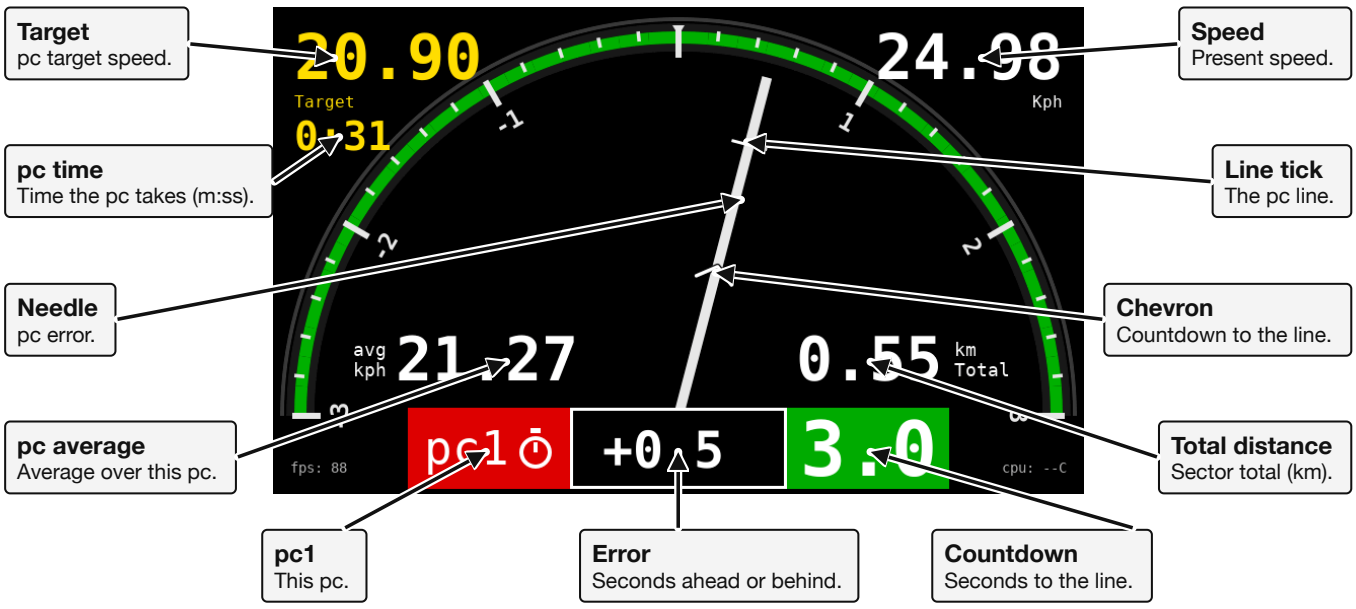
Two pages: one on navi, ct and co rows, and one inside a pc or pm. All figures are in km and kph.

Navigation (navi, ct and co rows)



The top left shows the row's target speed and, under it in yellow, how long the row takes at that speed (m:ss). There is no needle and the middle box is empty: on these rows crews drive flat out and a lead of many minutes is normal. The right-hand box shows the **minutes** in hand: green when ahead, red when behind.

In a pc or pm



Inside a pc or pm the needle and the error box give the **error** from the start tube, in seconds. The yellow time is how long the pc takes. The box on the right counts down the seconds to the line, from 5.0 to 0.0.

The chevron

Inside a pc or pm one chevron sits at the start of the needle, with the line tick further up. It waits there until 5 seconds are left to the line; then it slides up the needle to the tick, arriving as the countdown reads 0.0. It only does the countdown, and there is only one. On navi, ct and co rows there is no needle and no chevron.

SURREY RALLY TYPE

> **Not built yet:** the surrey key in settings is greyed. This section describes the agreed design.

For British road-book regularities such as the GoodBrook Surrey Hill Ramble. The roadbook gives one **running total** per section, from 0 at the section start, never reset at a box. A stage starts at a loosely given point in a box ("about 75 m after box 11"), runs over several boxes and finishes at a point in a known box, perhaps with no printed distance. The crew writes the **time taken** on the yellow sheet. Surrey works as greek except as noted here.

	greek	**surrey**
Readouts	**total** / **trip** / **countdown**	**total** (from the `now` press) / **trip** / **navi**, the roadbook's running total, zeroed only at a section start. countdown moves to the stage column.
Stage start	on a time (autostart) or `now`	`now` pressed at the landmark (autostart still works if a start time is entered)
Speed changes	at the row distance, `auto` or `next`	the same, at the roadbook total of the box
Stage end	on distance, or `next` on the last row	`next` pressed at the finish landmark; on distance if not pressed
Result	ahead / behind	ahead / behind while driving, then **time taken** (`9:20.6`)

Programming Surrey Stages

Select **surrey** rally from the settings page and then select `segments` from the main menu. The segments page will display a Union Jack and is otherwise as in Programming Greek Stages.

Entering Surrey stage data

Start time is usually blank: a surrey stage has no start minute.

1. **from** – the roadbook total of the box the start lies in (SS2: box 27, `22307`).
2. Each row's **to** – the roadbook total where the speed next changes, or the finish. Set speed changes after the first row to `auto`: box totals are known, so they happen by themselves.
3. If the finish has no printed distance, the last **to** is the end of its box, which is the next box's total (SS3: finish in box 62, last to `53897`, the total at box 63).

Stage	**from**	**rows (box, to, kph)**	**finish**
SS1	9148	11, 18566, 60	at box 23
SS2	22307	27, 24024, 65 · 30, 26752, 50 auto	at box 32
SS3	49732	59, 53897, 58	signpost in box 62

`upload stage`, `clear stage` and `clear all` work as in greek.

Starting a Surrey Stage

A surrey stage is started with `now` (Start Now) as the car passes the start landmark. Autostart still works if a start time is entered.

Driving a section

When	Press	What the box does
`Section start (box 1, box 66)`	`navi` › reset	navi = 0.
`At a box, if navi is out`	`navi` › set	navi set to the box figure (total unchanged). Always do this at the start box.
`Passing the start landmark`	`now`	Stage clock and distance start; navi carries on. Row 1 is shortened to end at its box, so later speed changes match the roadbook.
`A speed change at a box`	nothing (`auto`)	Changes speed at the box. `next` and `prev` work as in greek.
`Passing the finish landmark`	`next` on the last row	The stage ends; **time taken** shows under Trip history and on the phone.

Navigating a Surrey Stage

The navigation page

Rows: `total` (from the `now` press) / `trip` / `navi` (cyan, the roadbook running total; tap to set or reset). During a stage, **countdown** (metres and minutes to the next speed change or finish) sits at the foot of the stage column; between stages Stage arrival shows there. Under the Trip history (last four), **time taken** for the last stage stays until the next one finishes. `check total` shows in orange where `manual` normally is.

When things go wrong

Situation	What to do
navi total out at the start (a wrong turn before the stage)	Best: `set` the start box's figure as you pass it, then press `now`. Left uncorrected, the stage runs correctly from the press but automatic speed changes come at the wrong place: for each one, press `prev` if it comes early and `next` at the box. A large error (over 300 m before the start box figure, or past the end of row 1) also shows `check total`.
A wrong turn inside the stage	Correct **total** with `set` or `±10`, as in greek, and **navi** separately with `navi` › set at the next box.
`next` pressed too early at the finish	Press `prev`: the stage carries on as if nothing happened. Works until navi passes the last row's **to**.
Finish passed without pressing `next`	The stage ends by itself at the last row's **to**, as in greek, and time taken is to that point: right for a printed finish distance, but for a finish inside a box it is the time to the box end, not the landmark.
`now` pressed before the real start	As greek: press `now` again, confirm the abort, then press `now` at the real landmark. navi is unaffected.

The section target time (90 min for each section on the GoodBrook roadbook) is not shown.

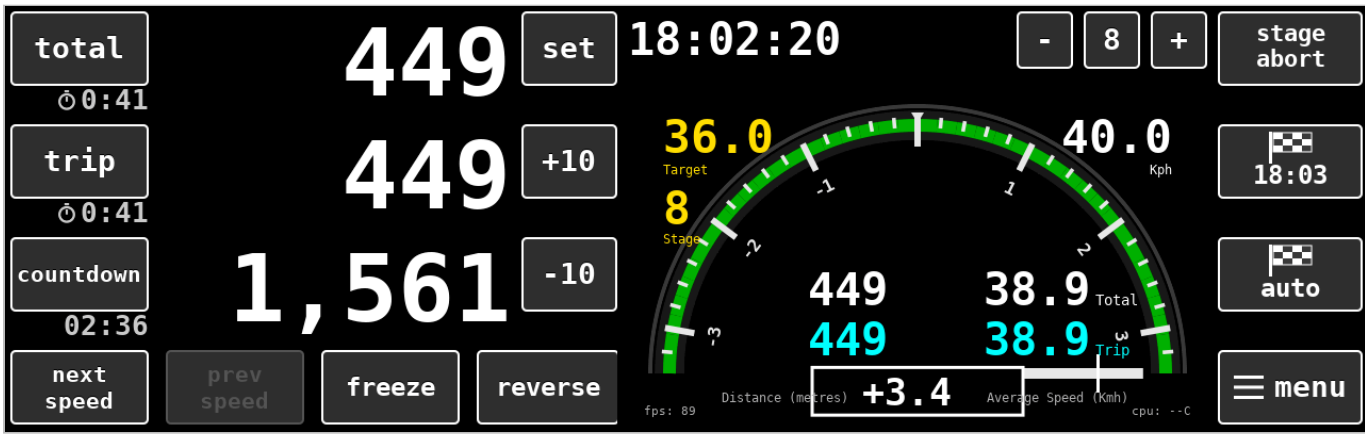
The driver display

The Surrey stage pages are the Greek ones: in a stage, the needle, error box and red **S** logo; between stages, the arrival speed, the **navi** logo and no needle.

The chevron

As in greek. One chevron rides the needle: it starts 20% up at the beginning of each row and slides up toward the row end line as the row is driven, arriving as the row ends. In Surrey a row ends at a box or at the finish, so the chevron counts down to the next box. Between stages there is no needle and no chevron.

SINGLE-DISPLAY MODE



With **Force single display** on, the driver's gauge shares the co-pilot screen. Keys work as on the navigation page. Left: Total / Trip / next with `+10`, `set` and `−10`; along the foot: `prev speed`, `freeze`, `reverse` and `A–B, B–C zero`; middle: rally clock and **– 1 +** stage selector above the gauge; right: the three flag keys and `menu`. No Trip history or stage table; other pages are reached through `menu`. A pending start's countdown shows over the gauge.

THE PHONE APP

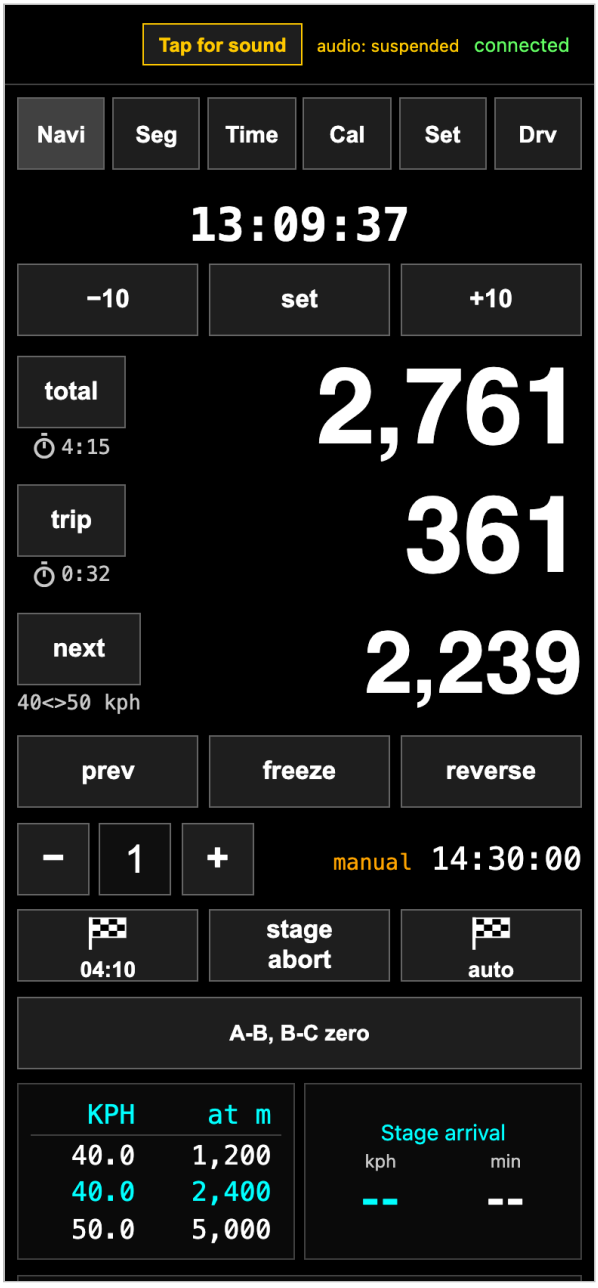
Connecting: link the box and phone over Wi-Fi, then open the address shown on the settings page (or scan its QR code). The page top reads **connected** in green when linked. Added to the phone's home screen, it runs full screen.

Sounds: the top key reads **Tap for sound** until the phone allows audio, then toggles Sound is on / off.

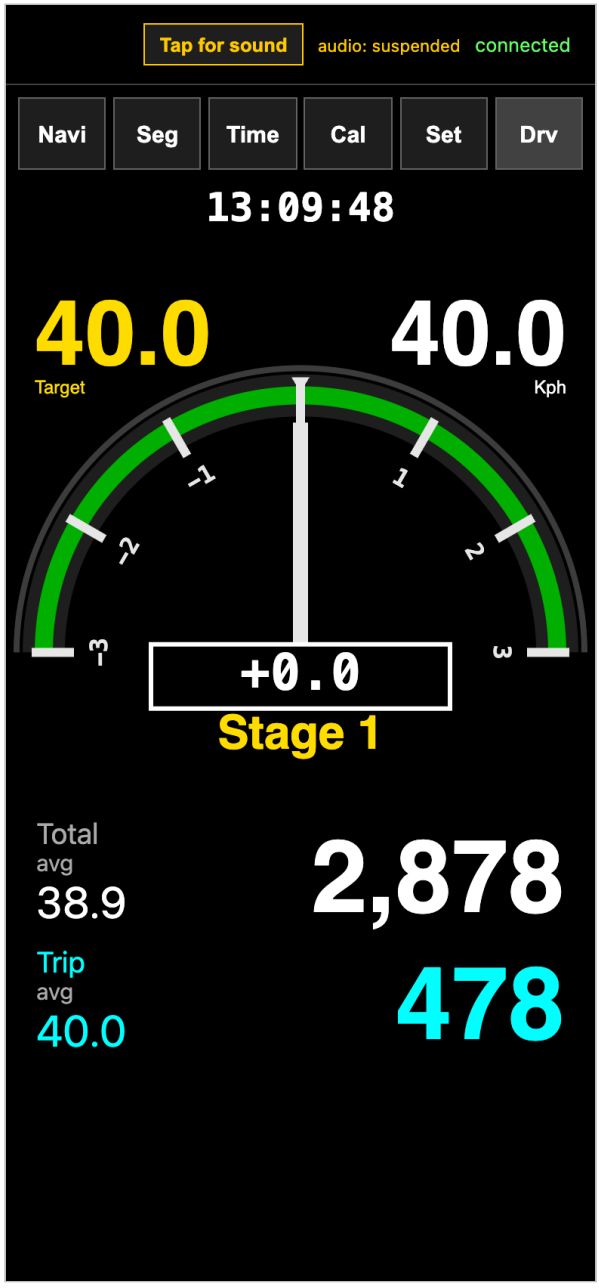
Nero2.1: the phone's stage selectors run 1–99 like the box; the phone does not yet show the italian columns, and its screenshots below are from 27 October (before Nero2.1).

Six tabs mirror the box: **Navi** (navigation, with total, trip, freeze, reverse, the three flag keys and A-B, B-C zero), **Seg** (segments), **Time** (date / time), **Cal** (calibration), **Set** (settings, plus `download logs`) and **Drv** (driver gauge with Total and Trip). On Navi and Drv, press Total or Trip to zero it.

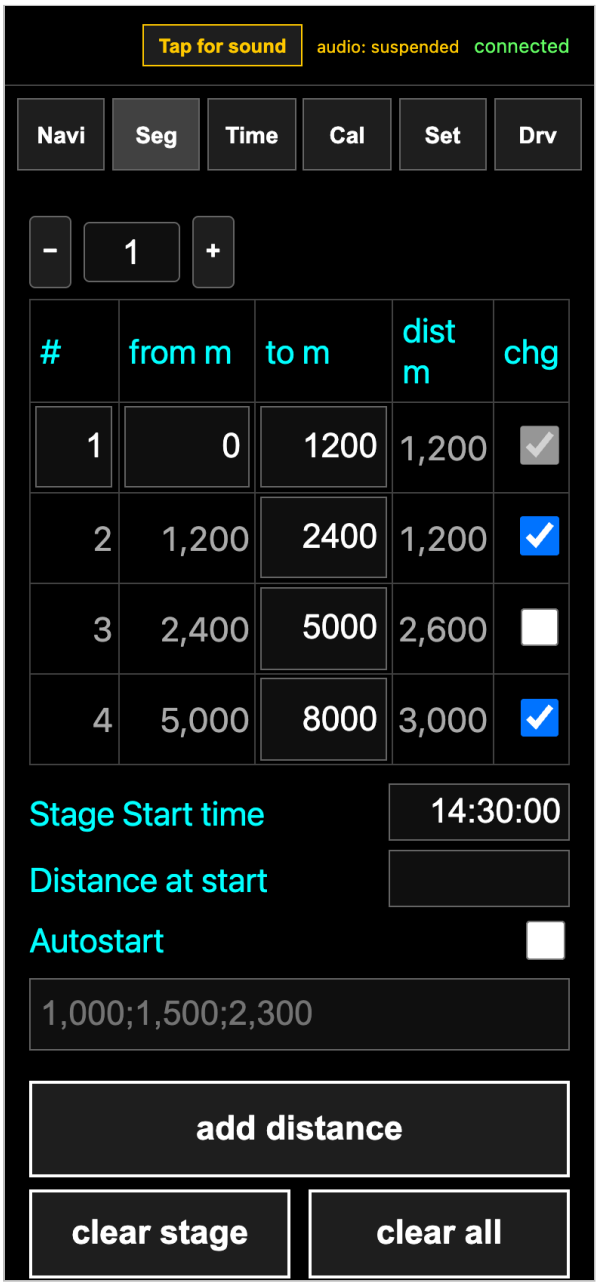
Navi



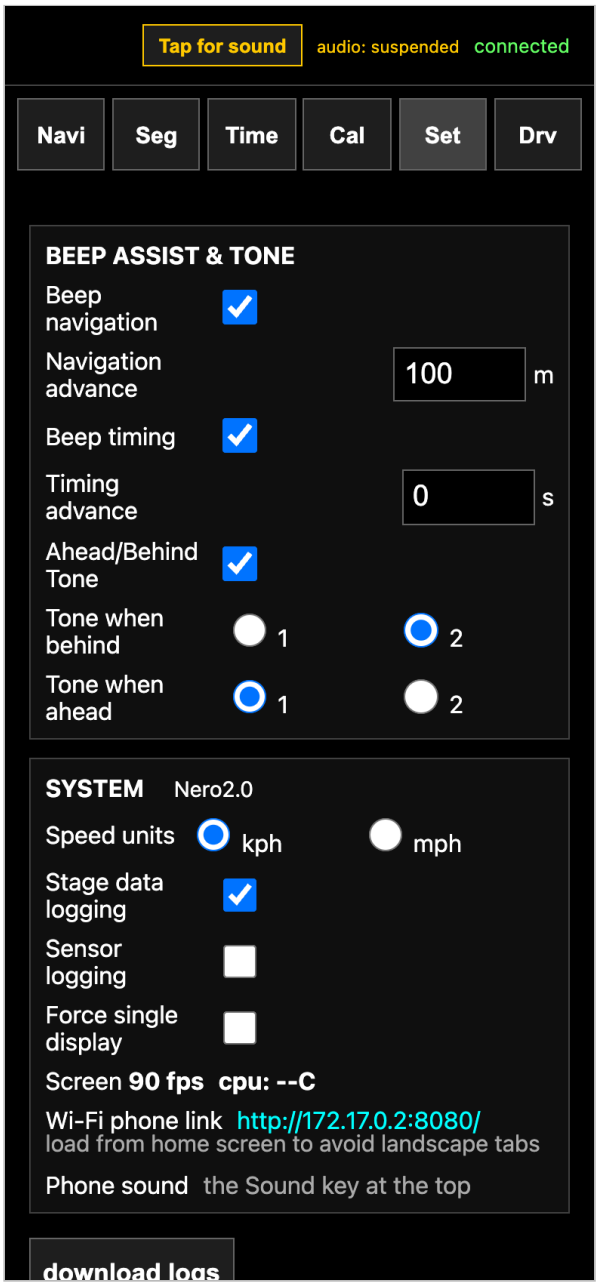
Drv



Seg


Tap for sound
audio: suspended
connected
Navi
Seg
Time
Cal
Set
Drv
-
1
+
from m to m dist m chg
1 0 1200 1,200
2 1,200 2400 1,200
3 2,400 5000 2,600
4 5,000 8000 3,000
Stage Start time
14:30:00
Distance at start
Autostart
1,000;1,500;2,300
add distance
clear stage
clear all

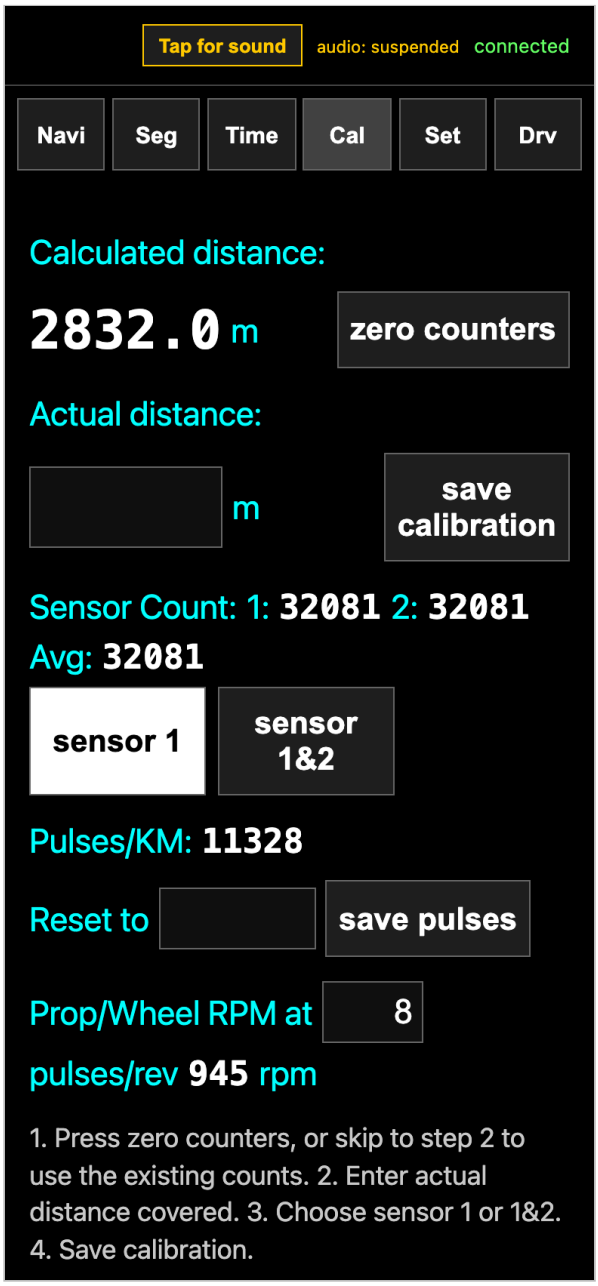
Set


Tap for sound
audio: suspended
connected
Navi
Seg
Time
Cal
Set
Drv
BEEP ASSIST & TONE
Beep navigation
Navigation advance
100
m
Beep timing
Timing advance
0
s
Ahead/Behind Tone
Tone when behind
1
2
Tone when ahead
1
2
SYSTEM
Nero2.0
Speed units
kph
mph
Stage data logging
Sensor logging
Force single display
Screen 90 fps cpu: --C
Wi-Fi phone link http://172.17.0.2:8080/
load from home screen to avoid landscape tabs
Phone sound the Sound key at the top
download logs

Time


Tap for sound
audio: suspended
connected
Navi
Seg
Time
Cal
Set
Drv
Rally time
13:09:42.2
−0.1
+0.1
Computer time
13:09:42.2
set rally to computer time
Phone time
13:09:42.2
set computer to phone time
Enter rally time
hh:mm:ss
set rally to entered time
Offset to computer
+0.0
Tue, 27 Oct 2026

Cal


Tap for sound
audio: suspended
connected
Navi
Seg
Time
Cal
Set
Drv
Calculated distance:
2832.0 m
zero counters
Actual distance:
m
save calibration
Sensor Count: 1: 32081 2: 32081
Avg: 32081
sensor 1
sensor 1&2
Pulses/KM: 11328
Reset to
save pulses
Prop/Wheel RPM at
8
pulses/rev 945 rpm
1. Press zero counters, or skip to step 2 to use the existing counts. 2. Enter actual distance covered. 3. Choose sensor 1 or 1&2. 4. Save calibration.

BEEP ASSIST

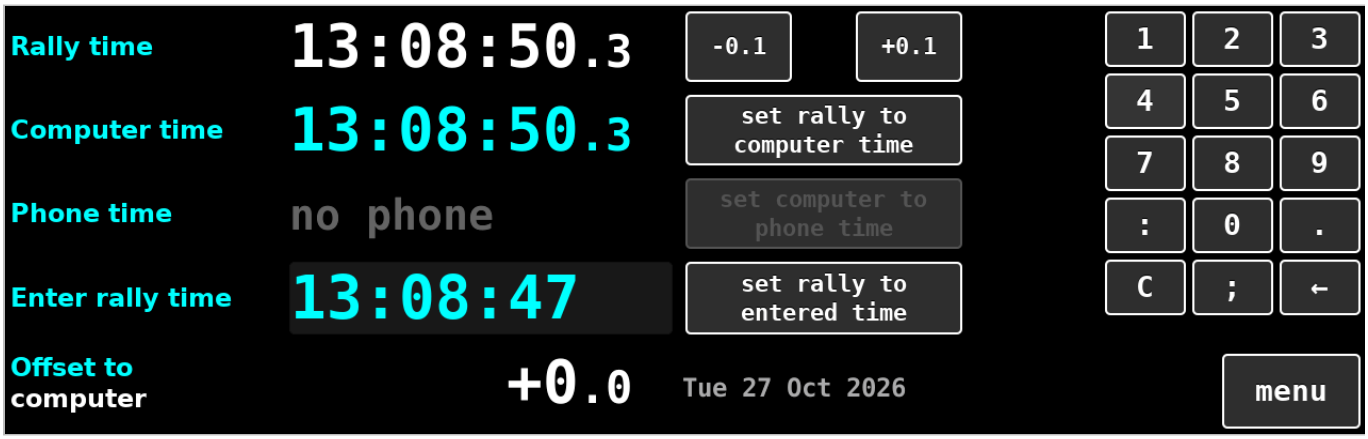
Beep Assist uses the segments **to** figures as the roadbook boxes; nothing more is entered. Switch it on in settings. Navigation double-beeps a set distance before each box. Timing single-beeps when the car should pass it, revealing distance errors from wheel spin, corner cutting or a skipping sensor. Beeps follow Total, so +10, -10 or set moves them all.

- **Beep navigation** – off/on. **Navigation advance**: metres before the box for the double beep (100 recommended).
- **Beep timing** – off/on. **Timing advance**: seconds before the box time for the single beep (0 recommended).

> Beep Assist is for greek stages and is silent in italian. There, inside a pc, the box instead beeps the countdown to the end tube: at 30 s and 15 s, then 5, 4, 3, 2, 1, and a short beep on 0.

ADJUSTING SETTINGS

date / time



The date / time page.

Rally time (white) is the clock all stages are timed against; **Computer time** (cyan) is the Raspberry Pi's clock; **Phone time** is the linked phone's, or `no phone`. **Offset to computer** is how far rally time is ahead (+) or behind (–), for example `+2.3`, or `–11:49.2` beyond a minute.

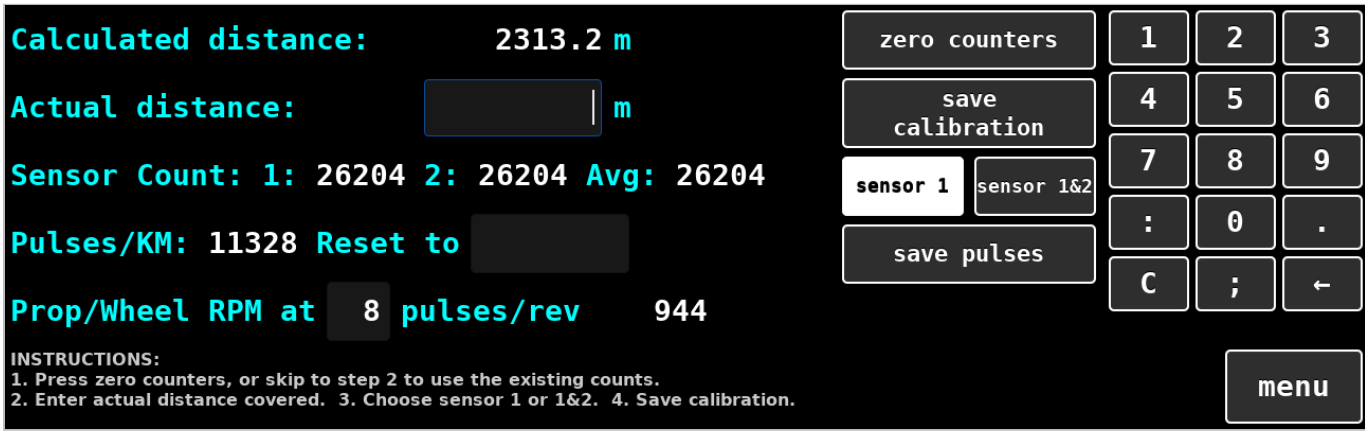
Setting the rally clock

- `set rally to computer time` makes them equal.
- Type `hh:mm:ss` (or `hh:mm`, no tenths) in **Enter rally time** and press `set rally to entered time`. Today's date is used.
- `set computer to phone time` sets the Pi's clock from the phone, leaving the rally clock as it is. Grey until a phone connects. It needs the Pi's usual password-free permission to set its clock; without it the box says `The computer clock could not be set` and nothing changes.

Trimming

`–0.1` / `+0.1` trim the rally clock by a tenth of a second; hold to repeat. Saved at once.

calibration



The calibration page.

Calibrating over a set distance

1. Press `zero counters` (or skip, to use the distance since the last Total reset).
2. Drive the measured distance (from 500 m to 100,000 m).
3. Type the **actual** distance driven (not what the box shows) in Actual distance.
4. Choose the counter: `sensor 1` or `sensor 1&2` (shown in white).
5. Press `save calibration`.

The page shows the calculated distance, each sensor's count and the pulses per km in use. If the distance is out of range or nothing was counted, `save calibration` silently does nothing, so check the pulses per km figure has changed.

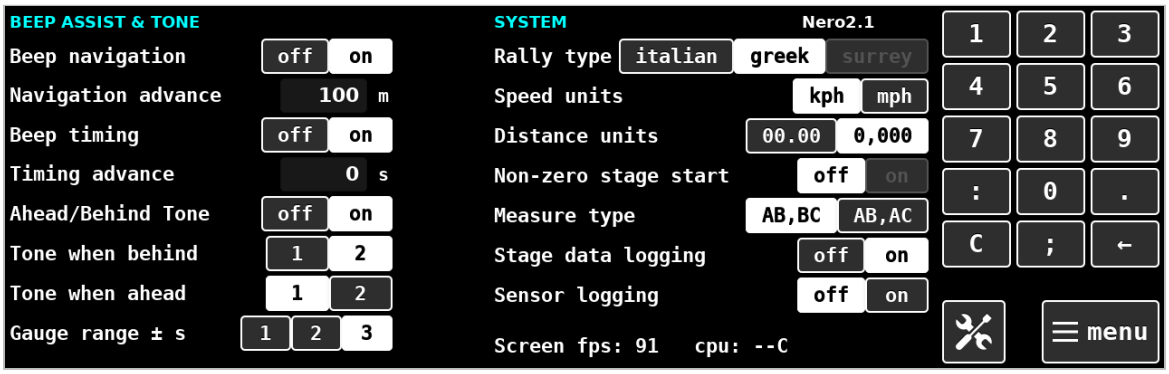
Setting the pulses directly

If the figure is known, type it in **Reset to** and press `save pulses`; no driving needed. Zero or less is ignored.

Prop/Wheel RPM

The last row shows propshaft or wheel rpm from sensor 1. Type the sensor's pulses per revolution in the small box.

settings



The settings page.

Each switch shows its present state lit. Changes are saved at once.

Tone

- **Ahead/Behind Tone** – the driver's error tone, off/on.
- **Tone when behind** / **Tone when ahead** – type 1 or 2; both silent under 0.2 s error and in the first 250 m. **Type 1**: a buzzer tick that speeds up with the error in 0.2 s steps, from about one a second at 0.2 s to about eight by 3 s; behind ticks slightly higher. **Type 2**: a steady tone rising in pitch in 0.2 s steps to 3 s; behind is higher and harsher, ahead lower and smoother.
- **Gauge range ± s** – 1, 2 or 3 seconds at each end of the gauge.

System

- **Rally type** – greek / surrey / italian. See Rally types. The software version (Nero2.1) is shown above it.
- **Speed units** – kph or mph.
- **Distance units** – **00.00** (km) or **0,000** (metres) for `add distance`.
- **Measure type** – **AB,BC** shows the A-B, B-C zero key on the greek navigation page; **AB,AC** hides it.
- **Stage data logging** – one file per stage, a row a second (distance, speed, ahead/behind error), from stage start to end or abort; switched on mid-stage, it starts at once. If the file cannot be written, logging silently switches off.
- **Sensor logging** – raw counter readings, many rows a second while on, so much larger files. Both logs are deleted after 60 days (at program start) and can be downloaded on the phone's Set tab.

The column foot shows the screen update rate and Pi temperature. The spanner key (keypad, bottom left) opens Setup.

Setup



The Setup page (in the sandbox no real displays are found).

- **Screen setup** – each display found, with `rotate` and a touch keyboard switch; `reset layout`; **Force single display** puts the gauge on the co-pilot screen from the next program start.

- **Bluetooth speaker setup** – `remember playing speaker` stores the speaker now playing; `connect` / `disconnect` it later.
- **Wi-Fi setup** – one key per saved network: the box's own **hotspot** (lit when used) or **join** another. Below: the phone app address and QR code. Opened from the phone's home screen, it runs without browser bars.
- `check for software updates` – closes the program and checks for an update (needs internet).
- `settings` returns to the settings page.

NOT BUILT YET

Agreed or planned, but the box does **not** do these today:

- `upload stage` on the segments page (greyed): loading a stage from a file.
- Italian: the count down **km** turning red and counting up past the line (the time already does).
- Italian driver display in km (it still reads *Distance (metres)*, Total and Trip).
- Beep Assist settings greyed in italian (it already stays silent there).
- **Reaction time offset** setting (all rally types).
- Italian columns on the phone's Seg tab; keeping phone edits made while disconnected.

THINGS TO CHECK

Seen while making this manual in the sandbox (30 October 2026). Each needs your judgement: is it what you want?

#	What was seen	Where
1	After **start now** (a manual flying start) the stage panel read **auto 18:01:56** – the word *auto* and the press time – not *manual*.	Greek navigation page, stage panel
2	The Trip history (101, 368) was empty once the stage had started.	Greek navigation page
3	Fixed 31 October: the greek Beep Assist no longer sounds in italian.	Italian, Beep timing on
4	Inside pc 1 the driver's gauge was marked to ±5 s although Gauge range was set to 3.	Italian driver display
5	The count down km stays at 0.00 once the line has gone by (a missed `next substage` does not show).	Italian navigation page
6	After the sector ended, the preview showed pc 1 as the line, not the opening navi substage.	Italian navigation page, idle
7	The word *manual* sits tight against a crossed-out start time.	Greek stage panel
8	Nothing in this manual has been tried on the real box, a phone, or in mph.	Everywhere

Suggested run-through

1. **Greek**: load a stage with Choose Stage (tap the stage number); start it each of the three ways; press `next speed` at a box; `prev speed`; +10 / set; freeze; abort. Check the countdown key's two figures against the segments page.
2. **Italian**: load sector 1; start in a minute and watch the +/– box; press `next substage` at the first tube and follow pc 1's count down to its end; press late at a tube and watch the time turn red; `prev substage`; drive a pm; finish at the co and compare the arrival time with the card.
3. **Settings**: switch rally type both ways and look at the segments and navigation pages; Measure type AB,BC / AB,AC; Gauge range.